ModelIt!

Student Activity Pack

# How Social Media Hacks Your Brain

Modeling Dopamine, Feedback Loops, and Digital Addiction

Grade 09L1 | NGSS HS-LS1-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

# Table of Contents

ModelIt!


1 Pre-Assessment ..... 3

2 Scoring Rubric ..... 4

3 Component Analysis ..... 5

4 Model Recording Page ..... 6

5 Simulation Observations ..... 7

6 Research & Extend ..... 8

7 Reflection Questions ..... 9

8 STEM Challenge ..... 10

# 1 Pre-Assessment

ModelIt!

**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **How many hours per day do you spend on social media, and how does it make you feel after using it?**

2. **Why do you think it's so hard to stop scrolling even when you know you should do something else?**

3. **What do you think happens in your brain when you get a like or comment on your post?**

4. **Do you think social media affects your mood or how you feel about yourself? Explain your reasoning.**

# 2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

ModelIt!

### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

Notification Frequency | Screen Time | Dopamine Release | Social Comparison | Self-Esteem Level

| EXTERNAL | INTERNAL |
|---|---|
| | |

### Explain your reasoning: Why did you sort each component the way you did?

# 4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Low Notifications

Notification Frequency set to low, simulating intentional phone use

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: High Notifications

Notification Frequency at maximum, simulating default app settings

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Gradual Escalation

Notification Frequency slowly increasing over weeks

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

ModelIt!

> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

### Possible New Components

**Sleep Quality** — The number of hours and depth of restorative sleep, which is severely disrupted by blue light exposure and dopamine-driven late-night scrolling, creating a secondary health cascade

**Anxiety Level** — The chronic state of worry and unease amplified by constant social comparison, fear of missing out (FOMO), and the pressure to maintain an online persona

**Algorithm Intensity** — The aggressiveness of the platform's recommendation algorithm, which learns what triggers dopamine release in each individual user and serves increasingly targeted content

### My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

ModelIt!

**The Big Question**

**Why can't you put your phone down — even when you know you should?** Use evidence from your simulation to support your answers.

1. **How does your model explain why social media companies design their apps with variable notification schedules rather than predictable ones?**

2. **What does the positive feedback loop in your model reveal about why quitting social media feels so difficult?**

3. **How does the relationship between social comparison and self-esteem create a trap that keeps users engaged?**

4. **If you could redesign one feature of social media based on your model, what would you change and why?**

5. **What are the ethical implications of companies knowingly engineering products that exploit the brain's reward system?**

# 8 STEM Challenge: Design a Digital Wellness App

ModelIt!

## YOUR ENGINEERING MISSION

Design an app or system that helps teenagers recognize and break the dopamine-driven feedback loop by providing real-time awareness of their usage patterns and emotional state.

### THE SCENARIO

A school district has reported that 73% of its students spend more than 4 hours daily on social media, with school counselors seeing a 40% increase in anxiety and depression referrals. The superintendent has hired your team to design a digital wellness tool that doesn't just limit screen time, but helps students understand WHY they keep going back and gives them strategies to break the cycle.

### Engineering Design Thinking

1. HOW WOULD YOUR APP DETECT THE DIFFERENCE BETWEEN INTENTIONAL USE (TEXTING A FRIEND) AND COMPULSIVE SCROLLING?

2. WHAT DATA VISUALIZATIONS WOULD HELP USERS SEE THEIR OWN DOPAMINE-DRIVEN PATTERNS?

3. HOW WOULD YOU INCORPORATE THE NEUROSCIENCE OF HABIT-BREAKING INTO YOUR APP'S INTERVENTION STRATEGIES?

4. HOW WOULD YOU TEST WHETHER YOUR APP ACTUALLY IMPROVES SELF-ESTEEM AND REDUCES COMPULSIVE USE?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

**! Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Behavioral Neuroscientists study how the brain drives behavior, including addiction and habit formation. They work in research labs, pharmaceutical companies, and tech ethics departments, earning $70,000–$140,000/year. UX Ethics Designers at companies like Apple and Google earn $90,000–$170,000/year designing technology that respects human well-being.